



University of Blida1. Faculty of Sciences / Computer Science Department
First-Year LMD: 2024/2025



ALGEBRA 1 EXAM. DURATION: 01H30. ANSWERS MUST BE BRIEF AND JUSTIFIED

Exercise (01) [4 points]: Using quantifiers and logical connectors translate the following propositions, then prove the validity of each proposition.

1. P_1 : "The product of any two rational numbers is rational".
2. P_2 : "For every integer n , if n^2 is even, then n is even".

Exercise (02) [4 points] : We consider the following application

$$f : \mathbb{R} - \{\ln(2)\} \longrightarrow \mathbb{R}$$

$$x \longmapsto f(x) = \frac{1 + 3e^x}{2 - e^x}.$$

1. Calculate $f(\{-\ln(2); 0\})$ and $f^{-1}(]-\infty, -3[)$.
2. Show that f is injective.
3. Determine the set $F \subset \mathbb{R}$ such that the application f is bijective from $\mathbb{R} - \{\ln(2)\}$ to F , then give the inverse application f^{-1} .

Exercise (03) [2.5 points]: In $\mathbb{R}$, we define the following binary relation

$$\forall x, y \in \mathbb{R}, x \mathcal{R} y \iff \left[\exists k \in \mathbb{Z} : e^{x+k} = e^y \right].$$

1. Show that $\mathcal{R}$ is an equivalence relation.
2. Determine $cl(0)$.

Exercise (04) [9.5 points]: Let $G_\beta = \mathbb{R} - \{\frac{-1}{\beta}\}$ where $\beta \in \mathbb{R}^*$. We defined on G_β an law denoted $*$ by

$$\forall x, y \in \mathbb{R} : x * y = \beta xy + x + y.$$

1. Show that $\forall x, y \in G_\beta, x * y \in G_\beta$.
2. Show that $(G_\beta, *)$ is an abelian group.
3. For $\beta = 1$. Let $f : (G_1, *) \longrightarrow (\mathbb{R}^*, \times)$ such that

$$f(x) = x + 1.$$

- a. Show that f is an homomorphism.
- b. For all $x \in G_1$, calculate the quantity

$$(x)^{(n)} = \underbrace{x * x * x \dots * x}_{n \text{ times}}, \forall n \in \mathbb{N}^*.$$

4. For $\beta > 0$, show that $H =]\frac{-1}{\beta}, +\infty[$, is a sub-group of $(G_\beta, *)$.